LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	30
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Flight data recorders, parameters selected methods of recording.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Flight data recorders, parameters selected methods of recording.
5. Specific Objectives	15.1.11.3 Trainee will be able to explain operation of Flight data recorder.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of FDR parameter lists from IB; Instructor supervision during Q&A; session.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of a Flight Data Recorder?
(b) Name two types of Flight Data Recorders used in modern aircraft.
(c) Where is the FDR typically located on an aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• FDR operational principles • Data acquisition systems • Parameter selection criteria • Recording media types • Black box identification features	L & Ds	PPT, IB	15 min
(b)	• Standard flight parameters • Environmental data recording • Control surface telemetry • Engine performance metrics • System status monitoring	L & Ds	PPT, PDF	20 min
(c)	• Data storage mechanisms • Impact protection standards • Recovery procedures • Data decoding basics • Regulatory compliance	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how parameters are selected for recording on an FDR.
(b) Describe the method used to protect FDR data during crash scenarios.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from analog to digital FDR systems for the next lesson on protection and location.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Protection and location of flight data recorder.	Refer to the master syllabus reference list	LP 31

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Engineer, P No. 12345	Senior Instructor, Chief Engineer, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min